

AWI-ESM3 on albedo: why the coupled runs are slow

Jan Streffing

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Abstract

The AWI-ESM3 CMIP7 spin-up (TCO95L91 OpenIFS, CORE3 FESOM2 with cavities, LPJ-GUESS, XIOS, 47 nodes) moved from levante to albedo on 2026-09-26. On albedo it runs at 49 to 57 SYPD, against 83 SYPD for the same configuration on levante and about 100 SYPD in Miguel Andrés-Martínez’s earlier albedo tests. Where we are: at the 1200s FESOM step FESOM limits the coupled model on albedo; at 1800s FESOM and OpenIFS are balanced on both machines, and albedo is about 1.1 times slower than levante in steady state. Returning FESOM to an 1800s time step brings albedo from 56 to 72 SYPD on the same 47 nodes. The remaining machine factor over levante is not explained by any MPI setting we could test: three transport and collective settings that make MPI benchmarks 2 to 7 times faster leave the model unchanged. A profile of FESOM on both machines shows its ranks polling in MPI for two thirds to three quarters of the time on levante as well as on albedo: FESOM at 1792 ranks is communication-bound everywhere, and albedo is 1.23 times slower in the same profiled run (Section 2). The timeline is in Section 1, falsified explanations are kept in Section 3.

1 The campaign, in the order it happened

1.1 First albedo legs (2026-09-26–27)

After the OASIS partition fix and the OpenIFS cold-start fix, the first clean 10-year legs on albedo took 4 h 20 min to 4 h 40 min of wall time (jobs 47456102, 47472034, 47472035). Levante had done the same legs in about 3 h. Four other legs died in an XIOS server with a SIGBUS; that problem is documented separately (`xios_sigbus_albedo_report_2026-09-28.md`) and is not part of the speed question.

1.2 Albedo against levante, same configuration (2026-09-28)

We compared the finished ccnice leg 2110–2119 on albedo (job 47472034) with levante’s leg 2090–2099 (job 27681023). Both used 47 nodes with FESOM 1792×1 , OpenIFS 384×4 , LPJ-GUESS 64×8 , XIOS 16×128 , identical namelists (`whichEVP=1`, $\alpha = \beta = 300$, `linfs`, `cavities`) and single-precision OpenIFS on both machines.

	levante	albedo	ratio
Leg wall time (s)	10385	15483	1.49
OpenIFS wall per step (s)	0.118	0.173	1.47
OpenIFS CPU per step, 4 threads (s)	0.452	0.548	1.21
FESOM ocean (s per leg)	4406	5694	1.29
vertical mixing	1111	1304	1.17
tracer advection	933	1128	1.21
ssh solver	1314	1850	1.41
FESOM ice EVP	2725	4992	1.83
FESOM waiting for the atmosphere (<code>ice read forc</code>)	441	1200	
FESOM output (means and XIOS send)	1925	1831	0.95

Table 1: Timings per 10-year leg. FESOM numbers are rank-0 means from its `MODEL_RUNTIME` per task block; OpenIFS from `ifs.stat` time stamps.

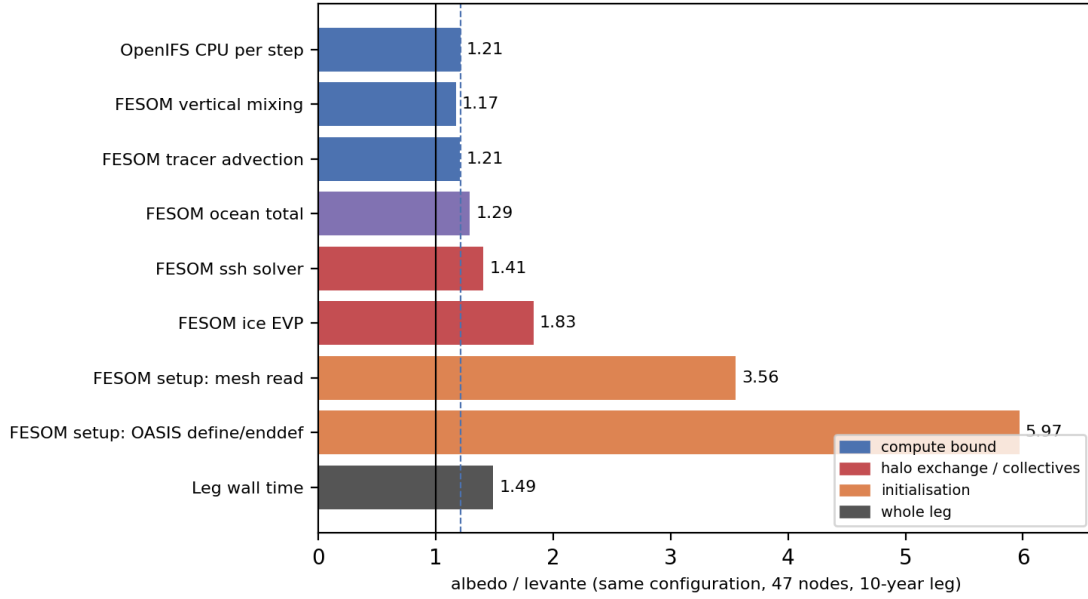


Figure 1: Albedo over levante for the same configuration. Compute-bound parts (blue) slow down by about 20 %, the dashed line; a cache-resident benchmark kernel measures levante’s Zen3 cores (EPYC 7763, 2449MHz under full load) as 16 % faster than albedo’s Zen2 cores (EPYC 7702, 2331MHz). Parts dominated by halo exchanges and collectives (red) slow down by 41 to 83 %, and the communication-heavy OASIS setup (orange) by a factor of six. The whole leg is 1.49 times slower.

The build is not the cause: both machines use the Intel compiler (ifort 2021.6 on albedo, 2021.5 on levante) with the same base flags, and albedo adds `-march=core-avx2 -qopt-malloc-options=2 -qopt-prefetch=5 -unroll-aggressive`, which levante does not have. Both machines have HDR100 InfiniBand.

1.3 Load balance on albedo from logs (2026-09-28)

This first estimate is superseded by LUCIA below. OpenIFS spends 0.143s of CPU per step on its 4 threads (0.571 over 4) out of a 0.173s step, so it computes 83 % of the time. FESOM is busy 98.5 % of the wall time when its `diag` (2.8 %) and `output` (12.1 %) categories are counted, of which 7.9 % is the wait for the atmosphere. Both models are saturated and FESOM is the marginally slower one. The daily LPJ-GUESS exchange costs 1.7 % (the step after it takes 0.242s against 0.17s), and there are no year-end stalls: every model year takes 1471 to 1645s.

1.4 Comparison with Miguel’s 100-SYPD runs (2026-09-28)

The albedo sheet of `AWIESM3-Performance.ods` lists 100.26 SYPD for FESOM 1792×2 , OpenIFS 384×4 on CORE3Cav, measured over steps 3–1416. Over the same window our leg runs at 53.2 SYPD. Two differences account for the gap:

- CMIP7 output. OpenIFS FullPos (DYNFPOS 16.9 %, GRIDFPOS 4.7 %, SU4FPOS 2.9 %) is 24.5 % of OpenIFS CPU for 209 files and 42 GB per model year. Without it the step would be about 0.108s. This output is required.
- FESOM at one thread. Miguel ran FESOM with two OpenMP threads, twice our FESOM cores, which removes the wait of OpenIFS on the ocean.

Miguel’s run directories are not readable for us (mode 700), so FESOM’s own timing in his runs is unknown.

1.5 XIOS is not a bottleneck (2026-09-28)

From the XIOS performance reports of the finished leg: the eight first-level servers are busy 2.6 to 2.8 % of the time, the eight second-level (file-writing) servers 34 to 41 %. The client ratio is 0.74 % on OpenIFS rank 0 and 0.04 % on FESOM rank 0. XIOS occupies 16 of 47 nodes with one server each, which is the one clear waste of node hours.

1.6 Initialisation is five times slower (2026-09-28)

A 1-day test leg (TEST_xios1d) takes 9 min: 5.5 min initialisation, 57 s of integration and 2.5 min for finalisation, tidy and the next leg’s preparation. File movement is threaded (127 threads) and takes about 20 s. FESOM’s setup timer is the dominant part:

FESOM setup on rank 0 (s)	levante (3 legs)	albedo (4 legs)	ratio
mesh and partition read	18–21	66–70	3.5
restart read	7–8	20–40	2.5
other: OASIS define and <code>oasis_enddef</code>	37–38	223–234	6
total	64–68	313–335	5

Table 2: The file reads slow down like GPFS I/O; the OASIS router setup, an all-to-all between 1792 and 384 ranks, slows down like the other communication-heavy parts. `oasis_enddef` also waits for the slowest model to finish its own initialisation, so part of it may be LPJ-GUESS or OpenIFS input on GPFS.

1.7 Node-to-node spread (2026-09-28)

The ccnice leg 2120–2129 (job 47473364) ran at 0.202 s per OpenIFS step (48.9 SYPD) against 0.173 s in the leg before, with the same configuration on a different set of nodes.

1.8 TEST_perfA: FESOM with two threads (2026-09-28)

A 2-month leg with FESOM 1792×2 (61 nodes) ran at 0.149 s per OpenIFS step (66.4 SYPD) against 0.176 s (56.0 SYPD) with one thread. FESOM’s compute per OpenIFS step fell only 8 % (0.145 to 0.134 s): the EVP by 23 %, the ocean by 7 %, while the ssh solver got 20 % slower. Doubling FESOM’s cores therefore buys little; FESOM on albedo is limited by memory bandwidth and communication, not by arithmetic. FESOM’s setup took 133 s in this run against 313 to 335 s in all earlier albedo runs (mesh read 22 against 66–72 s), with the same files on the same filesystem, so the initialisation cost varies from job to job.

1.9 LUCIA: FESOM is the limiter (2026-09-28)

With `oasis3mct.use_lucia: true` OASIS3-MCT 5 writes `load_balancing_info.txt` and one `timeline_<component>.nc` per component into the work directory. For TEST_perfA:

component	computing (s)	waiting (s)
FESOM	190.7	19.2
OpenIFS	92.8	125.4
LPJ-GUESS	≈ 0	300.9

Table 3: 2-month leg, 1416 OpenIFS steps. The OpenIFS numbers of this summary turned out to be wrong; see the timeline analysis later in this section, which gives 186 s computing and 54 s waiting. The CPU column of `ifs.stat` counts MPI busy-waiting as CPU time, which is why the log-based estimate above put OpenIFS at 83 %. The SYPD line of the LUCIA summary is broken (****); the compute and wait columns add up to the wall time and are sound.

1.10 What changed since August (2026-09-28)

The August albedo runs under `/albedo/work/user/jstreffi/runtime/awiesm3-develop/` (awiesm3sp.lucia and relatives, 1 month each, FESOM 1792×1 on the same CORE3 mesh with cavities) used a different FESOM configuration:

	August (awiesm3sp.lucia)	now (PICAL_ccnice)
time step	1800 s (48 per day)	1200 s (72 per day)
vertical mixing	KPP	cvmix TKE + IDEMIX
K_GM_max	1000	2500
FESOM compute per model hour	0.073 s	0.145 s
per FESOM step	0.037 s	0.048 s

Table 4: Same mesh, EVP settings (mEVP, 120 subcycles), ALE and rank count. FESOM now takes 1.5 times as many steps per day, each 32 % more expensive, which doubles its cost per model day. Since FESOM is the limiter, this accounts for most of the distance to the 100-SYPD runs in the performance sheet.

The August runs also show FESOM setup times between 14 and 274 s for identical configurations, which confirms that the initialisation cost is not a property of the configuration.

1.11 The MPI environment differs between the machines (2026-09-28)

esm_tools sets a tuned MPI environment on levante and almost none on albedo:

	levante (production job)	albedo (production job)
point-to-point layer	OMPI_MCA_pml=ucx, btl=self	OpenMPI default
collectives	HCOLL (coll_hcoll_enable=1, multicast)	OpenMPI tuned
transports	UCX_TLS=mm,knem,cma,dc_mlx5,dc_x,self, unified mode	UCX default
OpenMP runtime	KMP_AFFINITY=scatter, KMP_LIBRARY=turnaround	nothing
kernel	mitigations=off	all speculative-execution mitigations on

None of albedo’s OpenMPI builds (4.1.3, 4.1.5, 4.1.6) contains the HCOLL component, although libhcoll is installed under `/opt/mellanox/hcoll`. FESOM’s own `env/albedo/shell` builds with Intel MPI instead.

1.12 MPI and node benchmarks, albedo against levante (2026-09-28)

A small MPI benchmark (`bench/mpibench.c`) was compiled with each machine’s model MPI and run with 128 ranks per node:

	albedo, default	albedo, best setting	levante, production env
core speed, cache-resident kernel (GFlop/s per rank)	5.53		6.41
memory bandwidth, triad (GB/s per rank)	2.48		2.59
ping-pong 8 B between nodes (μ s)	1.6	1.4	3.4
allreduce 8 B, 1 node (μ s)	7.8		7.1
allreduce 8 B, 16 nodes (μ s)	50–88	21 (DC random)	30
halo 6×16 KiB, 1 node (μ s)	114	114 (default is best)	59
halo 6×2 KiB, 16 nodes (μ s)	164	23.5 (DC random)	
halo 6×16 KiB, 16 nodes (μ s)	287–343	148 (DC random)	74

Table 5: Per-core speed differs by 16 % and memory bandwidth by 4 %. Inside one node the halo exchange is twice as slow on albedo whatever transport is chosen (default, levante’s `mm,knem,cma,posix,sysv,CMA,knem`, no zero-copy, OpenMPI vader without UCX); the default is already the fastest. Across nodes, UCX’s DC transport with the default send policy `dcs_quota` is slow; `UCX_DC_MLX5_TX_POLICY=rand` makes collectives and small halos faster than levante. More DC initiators (16, 32, 64) or unified mode do not help. The benchmark runs used one leaf switch.

1.13 Model tests of the MPI settings and the time step (2026-09-28)

All tests are 2-month legs of the crunveg configuration on 47 nodes with FESOM 1792×1 and LUCIA on, run on `paleodyn.paleodyn`; the speed is measured over OpenIFS steps 3–1416.

test	MPI setting	FESOM step	s per OpenIFS step	SYPD
baseline (TEST_xiosbus)	default	1200 s	0.176	56.0
TEST_perfE (RC)	<code>UCX_TLS=rc_x,sm,self</code>	1200 s	fails at start-up	
TEST_perfE	<code>coll/han</code>	1200 s	0.184	53.6
TEST_perfG	DC, random policy	1200 s	0.188	52.4
TEST_perfF	<code>coll/han</code>	1800 s	0.137	71.9
TEST_perfH	DC, random policy	1800 s	0.136	72.6
levante production	levante env	1200 s	0.118	83.3

Table 6: The RC transport dies during initialisation with `ud_ep.c:280 Fatal: UD endpoint ... unhandled timeout error`: the OASIS all-to-all needs a connection per rank pair among 2256 ranks. Neither `coll/han` nor the DC random policy changes the model speed, although both make the benchmark collectives 2 to 7 times faster. Only the time step does.

FESOM per OpenIFS step (s)	baseline	<code>coll/han</code>	DC random
ice EVP	0.0576	0.0613	0.0558
ssh solver	0.0215	0.0230	0.0226
vertical mixing	0.0147	0.0148	0.0179
FESOM total (ice+ocean)	0.145	0.156	0.160

Table 7: The communication-heavy EVP and ssh solver do not move with the MPI settings. Even the purely local vertical mixing varies by 20 % between runs, which points at node-to-node variation. LUCIA shows FESOM computing 96 to 99 % of the time in all tests, so OpenIFS does not mask a FESOM gain.

At 1800 s FESOM costs 0.118 s per OpenIFS step instead of 0.156, the EVP 0.036 instead of 0.061. Both production runscripts were switched to 1800 s. The OASIS lags are all 3600 s and do not depend on it.

1.14 Earlier FESOM reports and the build (2026-09-28)

The FESOM tracker holds related albedo reports: #874 (FESOM 2.7 sea ice slow on every node on albedo, which went away after an albedo system update), #885 and #400 (`ENABLE_ALBEDO_INTELMPI_WORKAROUND` barriers in the native netCDF output gather against an Intel MPI message-queue pile-up; not relevant with OpenMPI and XIOS, and not in our branch), #1044/#1048 (barriers around the `sigma_xy` exchange cost 3–5 %; the fix is in our branch), #395 (`-mtune=core-avx2` for Zen, not set in our build) and #685 (GCC was twice as slow as Intel on levante; both our builds use Intel). Our build has the asynchronous I/O threads off (`ASYNCHRONOUS_IO_THREADS=OFF`), so no extra thread competes with the pinned FESOM ranks.

1.15 Profiling FESOM on albedo (2026-09-28)

VTune 2022.3 cannot run for normal users on albedo: `kernel.yama.pttrace_scope` is 2 and `perf_event_paranoid` is 2, and no sampling driver is loaded. We therefore wrote a small program-counter sampler (`bench/pcsampler.c`) that is preloaded into FESOM ranks 0, 896 and 1791 through `bin/fesom_prof.sh`. It samples the interrupted instruction pointer on a CPU-time timer (10 ms resolution at the kernel’s 100 Hz tick) and writes a histogram and the memory map at exit; `bench/pcs_report.py` turns them into time per library (FESOM, MPI, UCX, libc) and per FESOM routine. `TEST_prof` (2 months, 1800 s, default MPI) runs it. The first run sampled the whole job and was dominated by the initialisation, so the sampler now bins samples in 10 s intervals and the report is restricted to the time-stepping window taken from the `ifs.stat` time stamps. A wrapper for FESOM has to go through `fesom.execution_command`, because `bin_in_work` names the staged FESOM binary after `executable`.

The profile of the stepping window (250 s, about 2×10^5 samples per rank, job 47476015) is in Table 8. Two thirds of FESOM’s wall time is spent inside MPI. The UCX libraries are stripped, so their samples carry the name of the nearest exported symbol (`uct_dc_mlx5_ep_is_connected`, `uct_mm_iface_flush`); the code behind them is UCX’s static progress loops, which means FESOM ranks spend this time polling for messages, not doing work. The compute share differs between ranks: rank 896 computes 24 % of the time and has almost no sea ice (EVP 0.5 %), while ranks 0 and 1791 carry ice (EVP 4 to 5 %) and compute 30 to 35 %. The waiting is therefore at least partly the ice load imbalance, which the slower halo exchange on albedo makes more expensive. The accumulation of output means (`io_meandata::update_means`) costs 7 % of FESOM’s wall time on every rank, more than EVP and IDEMIX together and about a quarter of FESOM’s own computing; it does not depend on the machine. Thread-safety overhead of `MPI_THREAD_MULTIPLE` (`pthread_spin_lock`, `pthread_self`) is 8 to 10 %, but it sits inside the polling loop and would largely turn into more polling if removed.

share of wall time (%)	albedo (265 s)			levante (215 s)		
	r0	r896	r1791	r0	r896	r1791
FESOM code (<code>libfesom</code> , <code>fesom</code> , <code>libc</code>)	30.0	24.1	34.9	26.6	23.1	25.1
MPI, UCX and pthread (polling)	65.9	74.2	60.1	69.7	75.4	72.1
OpenMP runtime	3.9	1.7	4.9	3.5	1.3	2.7
<code>io_meandata::update_means</code>	7.5	7.1	7.0	7.2	7.0	7.1
<code>evpdynamics</code>	4.1	0.5	4.8	3.1	0.5	2.1
IDEMIX	1.9	1.8	2.6	1.8	1.7	1.8
<code>pthread_spin_lock</code> + <code>pthread_self</code>	6.1	10.2	7.6	5.3	6.1	6.4
FESOM code, seconds	79.5	63.9	92.5	57.2	49.7	54.0
MPI polling, seconds	174.6	196.6	159.3	149.9	162.1	155.0

Table 8: Program-counter profile of three FESOM ranks during the time stepping of the same 2-month run on both machines (`TEST_prof`, 2026-09-28, albedo job 47476015, levante job 27762013; stepping time in the header). The shape is the same on both machines: FESOM ranks compute a quarter to a third of the time and poll MPI for the rest, and the ice-free rank 896 waits longest. Albedo is 1.2 times slower in the polling and 1.1 to 1.7 times slower in FESOM’s own code.

The same run on levante (2 months from the same PICAL_ccnice restart, FESOM 1800s, levante’s default MPI environment and its leadclose FESOM binary) steps in 215s against albedo’s 265s, a factor of 1.23. Its profile has the same shape (Table 8): FESOM ranks poll MPI for 70 to 75 % of the time on levante too, `update_means` costs 7 % on both, and the thread-locking share is similar (5 to 6 % on levante). Albedo’s MPI is therefore not pathologically slow; FESOM at 1792 ranks is communication-bound on both machines, and albedo loses about 20 % in the communication and 10 to 70 % in FESOM’s own code. The largest code difference is on rank 1791, where EVP takes 12.7s on albedo against 4.5s on levante; the two FESOM binaries differ (albedo: main with PRs #1060 and #1087; levante: leadclose), so this part is not yet a pure machine comparison. The albedo sampler dropped 9 % of its samples when its hash table filled, levante’s under 1 %; the dropped samples fall preferentially on code with many distinct addresses, so albedo’s FESOM-code share is if anything underestimated. Both runs carry the sampler and LUCIA overhead; they are slower than production (albedo 53 against 72SYPD at 1800s) and are only compared with each other.

Why the factor is 1.23 here and 1.49 for the 10-year legs. Table 10 gives the stepping time of the first two months (1416 OpenIFS steps) in every run. Split by model day (1s resolution of `ifs.stat`), the two TEST_prof runs step equally fast between day 2 and day 58: 3.37s per model day on albedo, 3.35 on levante. The whole difference sits in two days. The first day, with the cold-started OpenIFS, takes 31s on albedo against 16s; the last day, when the leg writes its restarts and closes its output, takes 37s against 7s. In the 10-year legs at 1200s the steady state itself differs: 4.16s per model day on albedo (1517s per year) against 2.84 on levante (1036s per year), a factor of 1.46. LUCIA on levante (TEST_prof, 1800s) shows FESOM computing 178.7s and waiting 29.5s, OpenIFS computing 149.1s and waiting 63.1s: FESOM is the limiter on levante as well. Albedo’s OpenIFS line is again negative and cannot be used. The sampler slows the run (albedo 3.37s per model day with it against 3.03 in perfH without it), and it slows the three sampled FESOM ranks on which all others wait, so it may equalise the two machines artificially; a levante run at 1800s without the sampler settles it.

That run, TEST_1800 (levante job 27762423, TEST_prof without the sampler), steps 2.77s per model day between day 2 and day 58, against 3.03 for albedo’s perfH: a steady-state factor of 1.09 at 1800s, and 1.15 over the two months including the first and last day (182 against 210s). Levante gains only 2.5 % from the longer FESOM step (2.84s per model day at 1200s in the 2090 leg). LUCIA explains why: on levante at 1800s FESOM computes 146s and OpenIFS 149s, both wait about 31s, so the two components are balanced and OpenIFS now sets the pace as much as FESOM does. On albedo the LUCIA summary suggested FESOM still limits at 1800s; the timelines below show that it does not. The 1.46 of the 10-year legs is therefore a property of the 1200s FESOM step on albedo; at 1800s the remaining machine factor is about 1.1 in steady state.

1.16 LUCIA timelines: at 1800s both components are balanced on both machines (2026-09-28)

The LUCIA summary cannot be trusted for OpenIFS: its computing time is negative in most runs. The per-rank event timelines (`timeline.<component>.nc`) can: for every rank they hold the start and end of each GET and PUT. `bench/lucia_timeline.py` sums the time spent inside GET after the first coupling exchange (whose GET absorbs the other component’s initialisation) and counts the rest of the window as computing.

run	FESOM step	FESOM computing	FESOM waiting	OpenIFS computing	OpenIFS w
albedo perfE (han)	1200 s	294.0	2.9	203.5	
albedo perfG (DC random)	1200 s	290.7	10.4	201.8	
albedo perfF (han)	1800 s	214.3	16.3	202.9	
albedo perfH (DC random)	1800 s	208.8	19.8	209.3	
levante TEST_1800	1800 s	167.3	32.1	170.7	

Table 9: Seconds per 2-month leg, mean over ranks, from the LUCIA timelines. At 1200 s FESOM limits on albedo and OpenIFS waits 115 s. At 1800 s the two compute equally long on both machines, and the remaining waits on both sides come from steps in which one or the other is slower. Albedo is 1.22 times slower than levante in OpenIFS and 1.25 times in FESOM.

A faster FESOM alone therefore buys little at 1800 s: at most the time OpenIFS spends waiting (37 to 48 s of about 250 s), and in practice less, because in the other steps OpenIFS is the slower one. Further speed needs both components to become faster, or more cores for OpenIFS.

run	FESOM step	1 day	10 days	30 days	59 days
albedo ccnice leg 2110	1200 s	33	86	167	283
albedo ccnice leg 2120	1200 s	33	80	173	307
albedo perfE (han)	1200 s	33	71	155	291
albedo perfG (DC random)	1200 s	30	68	154	285
albedo perfF (han)	1800 s	32	61	122	233
albedo perfH (DC random)	1800 s	29	54	117	210
albedo TEST_prof (sampler)	1800 s	31	61	127	260
levante ccnice leg 2080	1200 s	15	39	102	212
levante ccnice leg 2090	1200 s	14	39	92	177
levante TEST_prof (sampler)	1800 s	16	42	107	214
levante TEST_1800	1800 s	15	43	99	182

Table 10: Cumulative stepping time in seconds after 1, 10, 30 and 59 model days, from the `ifs.stat` time stamps. The first model day (OpenIFS cold start) costs twice as long on albedo in every run.

1.17 FESOM main with the OpenMP performance stack (2026-09-29)

FESOM PRs #1081–#1085 and #1093–#1095 (node-owned gathers instead of locked scatters, `MPI_THREAD_FUNNELED`, no lock calls at one thread, EVP halo overlap) were merged into `main` on 2026-09-29. Neither machine’s coupled runs had them: albedo runs `leadclose-1087` (8de217d6), levante `fesom2-leadclose` (1753f0bf). `main` was merged into PR #1060 (`use_atm_ice_tskin`, which production uses), pushed as 1ef85546, and built with the production options. #1095 itself changes only `EVPdynamics` (`whichEVP=0`); we run `mEVP` (`EVPdynamics_m`), which #1094 does cover. Two 2-month tests repeat perfH (UCX DC random, FESOM 1800 s, LUCIA) with the new binary. perfJ runs FESOM at 1024 ranks $\times$ 2 threads on 16 nodes; its OASIS set is production’s (levante `dist_1792` order) reordered to `dist_1024` with `AWI-ESM/oasis_reorder_tool`, including grids, masks and areas, which the `CONSERV` couplings read. The order was checked against `nod2d.out` and the `my_list` files of both partitions.

run	FESOM	s per model day	FESOM computing	OpenIFS computing	FESOM
perfH, leadclose 8de217d6	1792×1 , 14 nodes	3.04	208.8	209.3	
perfI, main + #1060	1792×1 , 14 nodes	2.72	182.2	200.6	
perfJ, main + #1060	1024×2 , 16 nodes	2.74	160.7	177.6	
levante TEST_1800, leadclose	1792×1 , 14 nodes	2.77	167.3	170.7	

Table 11: Steady-state stepping (days 2–58) and LUCIA timeline seconds per 2-month leg. The new FESOM computes 13 % less at the same layout and 23 % less at 1024×2 ; albedo now steps as fast as levante did with the old binary. FESOM waits for OpenIFS in both new runs, so the coupled speed is set by OpenIFS; its computing time varies by about 10 % between runs on different nodes.

SST stays within 31.6 to 32.2°C maximum and -2.0°C minimum in both runs. After two months the root-mean-square SST difference is 0.67 K between perfH and perfI (same partition, different binary) and 0.73 K between perfI and perfJ (same binary, different partition), the size of the chaotic divergence; the global means agree within 0.08 K . The reordered coupling set therefore works end to end.

1.18 Fewer nodes: XIOS packed, FESOM and OpenIFS with more threads (2026-09-29)

XIOS is MPI-only; its 128 threads per server only reserved whole nodes for memory. The largest XIOS server reaches 44 to 46 GB in a 10-year leg (task 2249 in all three production legs) and 8.6 GB in two months, so four servers fit on a 257 GB node. Rank thread counts are kept at 2, 4 or 8, because albedo’s cores share L3 in groups of 4 and a 3-thread rank would straddle groups. New node-aware FESOM partitions `dist.384` ($n_{\text{part}} = 12, 32$) and `dist.768` ($12, 64$) were made with `fesom_meshpart` in a private copy of the mesh, and the coupling set was reordered to each with the same checks as for 1024. All runs use the new FESOM binary.

run	layout	nodes	s per model day	node-s per model day	FESOM computing	OpenIFS
perfI	reference	47	2.72	128	182.2	
perfK	XIOS 16×32	35	2.86	100	199.5	
perfL	perfK + FESOM 384×4	33	2.95	97	188.9	
perfM	perfK + FESOM 768×2	33	2.86	94	179.6	
perfN	perfK + OpenIFS 192×8	35	2.98	104	199.7	

Table 12: Steady-state stepping (days 2–58) and LUCIA timeline seconds per 2-month leg. perfM needs 26 % fewer node-seconds per model day than perfI at 5 % lower speed. OpenIFS at 192×8 is slower than at 384×4 . perfK is bit-identical to perfI, so its difference is placement and node noise only; OpenIFS computing time varies from 168 to 210 s between runs with the same OpenIFS layout, so single-run differences of 5 % are not significant.

XIOS servers were at most 15 % busy in perfK (24 % in perfI). SST stays within -2.1 and 32.7°C in all runs, and the runs with new partitions differ from perfI by 0.64 to 0.72 K RMS after two months, the chaotic spread. The XIOS memory growth over a full leg is not covered by two-month tests; a 10-year leg in the packed layout has to confirm that no server exceeds 64 GB.

2 State of the albedo performance (2026-09-28)

Proven:

- Albedo runs the same configuration 1.49 times slower than levante (15483 against 10385 s per 10-year leg).

- Albedo’s cores are 16 % slower on cache-resident work; memory bandwidth per core is within 4 %.
- Compute-bound parts are 17 to 29 % slower, consistent with the slower cores.
- Communication-bound parts are 41 % (ssh solver) and 83 % (ice EVP) slower, the OASIS setup six times.
- At 1200s FESOM limits the coupled model on albedo (LUCIA timelines: FESOM 294s, OpenIFS 204s computing per 2 months).
- FESOM’s cost per model day doubled since August: 1200s instead of 1800s time step, TKE+IDEMIX instead of KPP, K_{GM} 2500 instead of 1000.
- FESOM gains only 8 % from a second OpenMP thread.
- The FESOM time step of 1800s gives 72SYPD on 47 nodes against 56 at 1200s; both production runscripts now use 1800s.
- No MPI setting available on albedo speeds up the model: `coll/han` and the DC random send policy help the benchmark but not FESOM, RC fails at this size, and HCOLL is not built into any albedo OpenMPI.
- Inside one node the halo exchange is twice as slow as on levante with every transport available.
- The initialisation time varies between runs (FESOM setup 14 to 335s for identical configurations).
- CMIP7 output costs 24.5 % of OpenIFS CPU and is required.
- FESOM ranks spend 60 to 75 % of the stepping time polling in MPI and 23 to 35 % computing, on albedo and on levante alike; ice-free ranks wait longest. Output-mean accumulation costs 7 % of FESOM’s wall time on both (TEST_prof).
- The same 2-month profiling run steps 1.23 times slower on albedo than on levante at 1800s; the polling is 1.2 times slower, FESOM’s own code 1.1 to 1.7 times.
- In steady state the machine factor is 1.46 at 1200s (10-year legs) and 1.09 at 1800s (perfH 3.03 against TEST_1800 2.77s per model day). Over a 2-month leg it is 1.15, because the first model day (OpenIFS cold start) and the last (restart and output writing) take about twice as long on albedo.
- Levante gains only 2.5 % from the 1800s FESOM step, albedo 25 to 30 %. At 1200s FESOM limits on albedo (294 against 204s computing per 2 months, LUCIA timelines). At 1800s FESOM and OpenIFS compute equally long on both machines (albedo 209 and 209s, levante 167 and 171s), so a faster FESOM alone gains at most the 37 to 48s OpenIFS waits.
- OpenIFS is 1.22 times and FESOM 1.25 times slower on albedo than on levante in computing time.
- FESOM `main` with the OpenMP performance stack (#1060 merge 1ef85546) computes 13 % less at 1792×1 and 23 % less at 1024×2 ; albedo then steps at 2.72s per model day (87SYPD steady state), as fast as levante with the old binary, and OpenIFS becomes the limiter.
- With XIOS packed four servers per node and FESOM at 768×2 , the model runs on 33 instead of 47 nodes at 2.86s per model day (83SYPD), 26 % fewer node-hours per simulated year; OpenIFS at 192×8 is slower than 384×4 .
- XIOS servers are busy at most 41 %, so four servers per node are feasible.

Open:

- A standalone CORE3 FESOM run on both machines with identical setup would isolate the machine difference from the coupling; levante has CORE3 FESOM 2.6 runs to copy the setup from.
- Whether the production layout loses speed from spanning several leaf switches (**no-switch-enforce**); the benchmarks ran within one switch.
- Whether the intra-node halo penalty is the Zen2 core-complex layout (4 cores per L3 slice against 8 on Zen3); not tested directly.
- FESOM's timing in Miguel's 100-SYPD runs.
- Whether FESOM's polling is halo latency, ice load imbalance or the ssh solver's reductions; the sampler records no call stacks. A FESOM rank-count scan (fewer ranks, same layout otherwise) would show how far 1792 ranks are past FESOM's scaling limit.
- Why EVP on rank 1791 is 2.8 times slower on albedo; first rerun with the same FESOM binary on both machines.
- Cheaper output-mean accumulation in `io_meandata`, and a partition weighted for sea ice.

3 Falsified claims (kept, struck through)

- ~~OpenIFS computes 0.066 s and waits 0.089 s per step (LUCIA summary, TEST_perfA).~~ The summary's OpenIFS column is wrong (negative in most runs); the timelines give 186 s computing and 54 s waiting for perfA.
- ~~FESOM still limits on albedo at 1800 s.~~ Also from the LUCIA summary; the timelines show FESOM and OpenIFS computing 209 s each in perfH.
- ~~On levante FESOM does not limit at 1200 s, so the 1800 s step gains nothing there.~~ The evidence was a profiled levante run compared with unprofiled production legs. The unprofiled TEST_1800 later showed a small gain (2.5 %), with FESOM and OpenIFS balanced at 1800 s, so the conclusion was close but the argument did not hold.
- ~~FESOM computes only 84 % of the time and has slack, so its cores can be reduced.~~ The 84 % omitted FESOM's output and diagnostics categories; LUCIA shows FESOM computing 91 % and waiting 9 %.
- ~~OpenIFS computes 83 % of the time and both models are saturated.~~ The `ifs.stat` CPU column includes MPI busy-waiting; LUCIA shows OpenIFS waiting 57 % of the time for FESOM.
- ~~The initialisation is five times slower than on levante as a property of albedo.~~ The same files read in 22 s in TEST_perfA and FESOM setup ranged from 14 to 335 s in identical configurations; it varies from job to job.
- ~~Give the freed XIOS nodes to OpenIFS, 384 to 768 tasks.~~ OpenIFS scaling and stability end at about four times the truncation, 380 tasks for TCO95.
- ~~Tidy and file movement make the short legs slow.~~ Tidy is threaded and takes 20 s; initialisation takes 5.5 min.
- ~~A precision difference between the machines.~~ OpenIFS is single precision on both ("The model is using single precision").

- ~~Different FESOM build or namelists:~~ Namelists are identical and albedo's build targets AVX2 while levante's is generic.
- ~~The DC transport's default policy slows FESOM; UCX_TLS=rc_x,sm,self or UCX_DC_MLX5_TX_POLICY=rand fixes it.~~ Both speed up the benchmark by 2 to 7 times, but RC fails at 2256 ranks and the DC random policy leaves FESOM's EVP and ssh solver unchanged (TEST_perfG/H).
- ~~Hierarchical collectives (coll/han) close the allreduce gap for the ssh solver.~~ Benchmark allreduce 53 to 24 μ s, model unchanged (TEST_perfE/F).
- ~~Kernel mitigations make intra-node exchange slow through system calls in CMA/knem.~~ Transports without system calls (posix, sysv, vader) are as slow or slower than the default.
- ~~Albedo's cores are 1.5 to 1.8 times slower on cache-resident code.~~ Measured 16%.
- ~~A GCC build or missing optimisation flags (FESOM PR #685).~~ Both machines build with Intel ifort; albedo has more aggressive flags than levante.

4 Method notes

- OpenIFS wall per step from time stamp differences in `ifs.stat`; CPU per step is column 5.
- OpenIFS routine shares from the timing statistics at the end of `NODE.001-01`.
- FESOM: the `MODEL RUNTIME per task` and `runtime setup` blocks of rank 0 in the compute log.
- XIOS: `Performance report` : Ratio at the end of `xios_server*.out` and `xios_client*.out`.
- LUCIA is switched on by `oasis3mct.use_lucia: true`, which writes `$NLOGPRT 1 0 1` for OASIS3-MCT 5. The result is `load.balancing_info.txt` and `timeline_*.nc` in the work directory; `oasis/util/load.balancing/pyLucia.py` plots the timelines.
- The CPU column of `ifs.stat` includes MPI busy-waiting and must not be read as OpenIFS's compute share.
- Short tests on albedo: `compute_time` of 30 min or less puts the job into the 30min QOS (priority 1 instead of 0).
- Standalone runs that set `--cpu.bind=none` with several OpenMP threads per rank and no `taskset` leave ranks and threads unpinned; the FESOM 2.8 release tests on 18 nodes collapsed that way (ssh solver 40 to 77 times slower than on one pinned node).
- Profiling: on levante the sampler's timer signals during start-up broke XIOS's collective `nc_create_par` for the interpolation weights ("Permission denied" on all FESOM ranks, job 27761837); `PCS_DELAY=60` starts the timer after the files are written. `PCS_ENABLE=1` and `LD_PRELOAD=libpcsampler.so` on selected ranks; `pcs_report.py <base> [top] [t0 t1]` restricts to an epoch window. Symbols of stripped libraries (UCX) are only the nearest exported name.
- Benchmark: `bench/mpibench.c`; job scripts `bench/bench_*.slurm`; the `HALO_BYTES` variable sets the halo message size.

5 Assets

- Runtime: `/albedo/work/projects/p_awiesm3_cmip7/runtime/awiesm3-v3.4/`, experiments `PICAL_ccnice`, `PICAL_crunveg`, `TEST_xiosbus`, `TEST_perfA-H`, `TEST_prof`, `TEST_xios1d`; failed attempts kept as `*_rc_failed`, `*_knobfail`, `*_vtune_ptrace`.
- Benchmarks and profiler: `esm_tools/albedo_performance_report/bench/`; levante copies in `/scratch/a/a270092/lr_probe/`.
- Profiling wrapper and sampler: `model_codes/awiesm3-develop/bin/fesom_prof.sh`, `libpcsampler.so`.
- Runscripts: `esm_tools/runscripts/awiesm3/develop/awiesm3-develop-albedo-TC095L91-CORE3*.ya`.
- Levante reference: `/work/bb1469/a270092/runtime/awiesm3-v3.4/PICAL_ccnice/`.